

Betty Wen

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PROFESSIONAL SUMMARY

University of Toronto

Bachelor of Applied Science in Mechanical Engineering + PEY Co-op

Sep 2022 – present

Toronto, ON

Technical Skills: SolidWorks, AutoCAD, PDM, MBD, Autodesk, Python, ANSYS, COMSOL, LTSpice, MATLAB, Fusion 360, Minitab

Related Courses:

WORK EXPERIENCE

Thermal and Mechanical Engineer

Litens Automotive Group

May 2025 – May 2026

Ontario, Canada

- Designed and optimized EV thermal and mechanical components, with a focus on 3D printing and rapid prototyping.
- Gained hands-on experience designing, assembling, and validating nine 70:1 humanoid cycloidal actuators across three design iterations for testing and demonstration, followed by torque-speed characterization and market benchmarking.
- Created GD&T drawings and integrated precision measurement data into 3D models for different purposes
- Developed a single-plate test rig supporting 3 Nm deadweight calibration and 40 Nm electrical gear-train testing.
- Performed FEA Analysis under varying torque and load conditions to evaluate stress concentrations and recommend gear materials.

Mechanical Design Engineering Intern

Tetrabot

May 2026 – August 2026

China

- Redesigned a ceiling-mounted robotic EV charging robots through three optimization iterations, improving its appearance, installation practicality, and compatibility with the existing rail and cable-reel mechanisms.
- Developed a spring-loaded traction module using the existing mounting holes, integrating a rubber-coated bearing and approximately 30 N of preload to improve wheel–rail contact and reduce slipping.
- Produced SolidWorks models, system renderings, and manufacturing documentation, incorporating feedback from engineers, mentor-guided design reviews, and factory visits.

Structural Engineer

Yuwell Medical Devices

June 2024 – August 2024

China

- Modeled and designed vibration-damping and ventilation components for oxygen concentrator compressors using CAD.
- Utilized 3D printing to prototype the designed module and conducted performance tests on different points of the module, successfully maintaining the overall compressor temperature below 55 Celsius degrees for over 1.5 hours.
- Acquired hands-on expertise with oxygen concentrators, emphasizing the enhancement of component longevity, thermal efficiency, heat management, and manufacturability

PROJECT

Mechanical Engineering Capstone Project – VLA-Based Robotic Tasks

University of Toronto

September 2026 – Present

Toronto, Canada

- Applying ML techniques, including transfer learning, fine-tuning, and validation, to adapt the pretrained $\pi_{0.5}$ VLA model for robotic bed-making and dishwashing across varied configurations.
- Building a task-specific demonstration dataset containing synchronized multi-view images, robot states, action sequences, language instructions, and success labels for model training and quantitative performance evaluation.

Undergraduate Research Assistant

NUS Multiphase Oil-Water-Air Flow Loop Laboratory, National University of Singapore

May 2024 - June 2024

Singapore

- Conducted CFD simulations of gasoline–air multiphase flow using ANSYS and COMSOL, applying the VOF method to capture phase-interface evolution and wavy annular flow and to analyze pressure, velocity, and phase distributions.
- Performed FEA to evaluate pipeline stability and structural response, integrated the CFD and FEA findings, and authored a 12+ page research paper proposing improvements to two-phase flow safety and efficiency.